

Appendix 1: Illustrative Example of CLEAR

This appendix provides a small numerical example to clarify how CLEAR transforms raw HDFS observations into category-aware replication and locality decisions. The example is synthetic; its purpose is not to report experimental performance, but to illustrate the successive computations used by the controller.

Eight files $f_1, \dots, f_8$ stored in HDFS are observed during one rolling window P . During this window, the Log Ingestor and the Feature Builder collect the raw counters needed to compute the five signals used by CLEAR: access frequency, file age, write ratio, locality, and peak concurrency. Table 1 summarizes these counters.

Table 1: Raw usage counters over one window P

File	Total accesses	Writes	Age (days)	Local accesses	Remote accesses	Peak conc
f_1	900	160	2	780	120	15
f_2	800	120	3	700	100	12
f_3	760	80	10	228	532	13
f_4	700	40	12	191	509	12
f_5	320	32	20	230	90	5
f_6	300	24	25	225	75	4
f_7	40	8	90	38	2	1
f_8	30	16	120	28	2	1

0.0.1 Step 1: From Raw Logs to Normalized Features

The Feature Builder converts the raw counters into the five features used by the clustering stage. The access frequency $freq(f_i)$ is the total number of accesses during P , the age $age(f_i)$ is measured in days since creation, and locality is the fraction of accesses served locally.

The global mean number of writes is

$$\mu_{\text{writes}} = \frac{1}{8} (160 + 120 + 80 + 40 + 32 + 24 + 8 + 16) = 60.$$

The write ratio is therefore

$$wRatio(f_i) = \frac{writes(f_i)}{\mu_{\text{writes}}}.$$

Table 2 reports the resulting feature values before normalization.

Table 2: Five per-file features before normalization

File	$freq(f_i)$	$age(f_i)$ (days)	$wRatio(f_i)$	$locality(f_i)$	$conc(f_i)$
f_1	900	2	2.67	0.87	15
f_2	800	3	2.00	0.88	12
f_3	760	10	1.33	0.30	13
f_4	700	12	0.67	0.27	12
f_5	320	20	0.53	0.72	5
f_6	300	25	0.40	0.75	4
f_7	40	90	0.13	0.95	1
f_8	30	120	0.27	0.93	1

Min-max normalization is then applied feature by feature to map all five dimensions to $[0, 1]$:

$$x'_{ip} = \frac{x_{ip} - \min(x_p)}{\max(x_p) - \min(x_p)},$$

with $x'_{ip} = 0$ when $\max(x_p) = \min(x_p)$.

Table 3 shows the resulting normalized feature vectors.

Table 3: Normalized features $\mathbf{x}'_i$ in $[0, 1]^5$

File	$freq'$	age'	$wRatio'$	$locality'$	$conc'$
f_1	1.000	0.000	1.000	0.877	1.000
f_2	0.885	0.008	0.737	0.889	0.786
f_3	0.839	0.068	0.474	0.040	0.857
f_4	0.770	0.085	0.211	0.000	0.786
f_5	0.333	0.153	0.158	0.658	0.286
f_6	0.310	0.195	0.105	0.705	0.214
f_7	0.011	0.746	0.000	1.000	0.000
f_8	0.000	1.000	0.053	0.975	0.000

0.0.2 Step 2: Clustering in the Feature Space

The clustering stage applies X-Means with K-means++ initialization to the normalized feature vectors. The search starts from $K_{\min} = 4$ and is bounded by $K_{\max}$.

For this synthetic example, the final partition contains four clusters:

$$C_1 = \{f_1, f_2\}, \quad C_2 = \{f_3, f_4\}, \quad C_3 = \{f_5, f_6\}, \quad C_4 = \{f_7, f_8\}.$$

Files f_1 and f_2 have very high access frequency, high write ratios, and strong concurrency. Files f_3 and f_4 also show sustained frequency and concurrency, but their locality is low because a large share of accesses is remote. Files f_5 and f_6 remain closer to the centre of the workload. Files f_7 and f_8 are old, rarely accessed, and almost inactive.

Table 4: Clusters and qualitative behavior

Cluster	Files	Qualitative profile
C_1	$\{f_1, f_2\}$	Very high frequency, high write ratio, high concurrency
C_2	$\{f_3, f_4\}$	High frequency and concurrency, very low locality
C_3	$\{f_5, f_6\}$	Moderate frequency and concurrency, near the global centre
C_4	$\{f_7, f_8\}$	Old files, very low frequency and concurrency

0.0.3 Step 3: Category Scores and Labels

For each feature p , CLEAR computes the global median $M_{\text{global},p}$ and the median absolute deviation MAD_p from the normalized values in the current window. Table 5 reports these values.

Table 5: Global medians and MAD scales per feature

Feature p	$M_{\text{global},p}$	MAD_p
freq'	0.552	0.310
age'	0.119	0.093
wRatio'	0.184	0.158
$\text{locality}'$	0.791	0.158
conc'	0.536	0.321

For example, the normalized write ratios are

$$1.000, 0.737, 0.474, 0.211, 0.158, 0.105, 0.000, 0.053.$$

After sorting,

$$0.000, 0.053, 0.105, 0.158, 0.211, 0.474, 0.737, 1.000,$$

and the global median is

$$M_{\text{global},wRatio} = \frac{0.158 + 0.211}{2} = 0.184.$$

The median absolute deviation is

$$\text{MAD}_{wRatio} = 0.158.$$

For each cluster C_k and feature p , the cluster median $M_{k,p}$, signed deviation $\Delta_{k,p}$, and standardized gap $z_{k,p}$ are then computed as

$$\Delta_{k,p} = M_{k,p} - M_{\text{global},p}, \quad z_{k,p} = \frac{\Delta_{k,p}}{\text{MAD}_p + \epsilon_0}.$$

Table 6 summarizes the standardized gaps, rounded to two decimals.

Table 6: Standardized gaps $z_{k,p}$ per cluster and feature

Cluster	freq'	age'	wRatio'	$\text{locality}'$	conc'
C_1	+1.26	-1.23	+4.33	+0.58	+1.11
C_2	+0.81	-0.45	+1.00	-4.86	+0.89
C_3	-0.74	+0.59	-0.33	-0.69	-0.89
C_4	-1.76	+8.09	-1.00	+1.24	-1.67

For C_1 , the cluster median of the normalized write ratio is

$$M_{1,wRatio} = \frac{1.000 + 0.737}{2} = 0.868,$$

so

$$\Delta_{1,wRatio} = 0.868 - 0.184 = 0.684, \quad z_{1,wRatio} = \frac{0.684}{0.158} \approx 4.33.$$

Bounded contribution functions. To make the category scores directly comparable, all contributions are mapped to $[0, 1]$. For features expected above the global level,

$$g^+(z) = \frac{\max(z, 0)}{1 + \max(z, 0)}.$$

For features expected below the global level,

$$g^-(z) = \frac{\max(-z, 0)}{1 + \max(-z, 0)}.$$

For the *Moderate* category, proximity to the global median is measured by

$$g^0(z) = \frac{1}{1 + |z|}.$$

With equal per-feature weights, the directional category score is

$$S_c^{(k)} = \sum_{p \in \mathcal{H}_c} w(p, c) g^+(z_{k,p}) + \sum_{p \in \mathcal{L}_c} w(p, c) g^-(z_{k,p}),$$

whereas the Moderate score is

$$S_{\text{moderate}}^{(k)} = \sum_{p \in F_{\text{moderate}}} w(p, \text{moderate}) g^0(z_{k,p}).$$

For example, the Hot score of C_1 is

$$\begin{aligned} S_{\text{hot}}^{(1)} &= \frac{1}{3} \left(\frac{1.26}{1 + 1.26} + \frac{4.33}{1 + 4.33} + \frac{1.11}{1 + 1.11} \right) \\ &\approx 0.63. \end{aligned}$$

Similarly, the Shared score of C_2 combines high frequency, low locality, and high concurrency:

$$\begin{aligned} S_{\text{shared}}^{(2)} &= \frac{1}{3} \left(g^+(0.81) + g^-(-4.86) + g^+(0.89) \right) \\ &\approx 0.58. \end{aligned}$$

Table 7 reports the four category scores for each cluster.

Table 7: Category scores and assigned labels

Cluster	$S_{\text{hot}}^{(k)}$	$S_{\text{shared}}^{(k)}$	$S_{\text{moderate}}^{(k)}$	$S_{\text{archival}}^{(k)}$	Assigned label
C_1	0.63	0.36	0.46	0.00	Hot
C_2	0.47	0.58	0.54	0.00	Shared
C_3	0.00	0.14	0.55	0.38	Moderate
C_4	0.00	0.00	0.37	0.66	Archival

The label is selected by

$$L_k = \arg \max_{c \in \{\text{Hot}, \text{Shared}, \text{Moderate}, \text{Archival}\}} S_c^{(k)}.$$

The bounded formulation keeps all four scores on the same scale and removes the need for category-specific score thresholds in this step.

0.0.4 Step 4: Replication Factors per Cluster

The RF Decision Module converts the assigned labels and their confidence scores into replication levels. The category-specific bounds used in this example are shown in Table 8.

Table 8: Replication bounds per category

Category	$R_{\min}(c)$	$R_{\max}(c)$
<i>Hot</i>	3	4
<i>Shared</i>	3	3
<i>Moderate</i>	2	3
<i>Archival</i>	0	1

For each cluster, the confidence score is the score of its assigned category:

$$s_k^* = S_{c_k}^{(k)}.$$

For a category c , let $\mathcal{K}_c$ be the set of cluster indices assigned to that category. Strong fit is allowed only when

$$|\mathcal{K}_c| \geq 3 \quad \text{and} \quad \text{MAD}_c > 0.$$

When these support conditions hold, cluster C_k is a Strong fit if

$$s_k^* \geq \text{med}_{c_k} + \text{MAD}_{c_k}.$$

Otherwise, the Strong fit flag is **false** and the minimum category-specific replication level is retained.

The toy partition contains one cluster per category. Consequently, $|\mathcal{K}_c| = 1$ for every category and the minimum-support condition is not satisfied. All four Strong fit flags are therefore **false**. This behavior is intentional: the safeguard prevents an isolated cluster from being promoted only because it is the sole representative of its category.

Table 9: Category, confidence, Strong fit flag, and replication decision

Cluster	Files	Assigned category	s_k^*	Strong fit	Replication decision
C_1	$\{f_1, f_2\}$	Hot	0.63	false	$R_1 = 3$ full replicas
C_2	$\{f_3, f_4\}$	Shared	0.58	false	$R_2 = 3$ full replicas
C_3	$\{f_5, f_6\}$	Moderate	0.55	false	$R_3 = 2$ full replicas
C_4	$\{f_7, f_8\}$	Archival	0.66	false	RS(6,3) EC only

For Hot, Shared, and Moderate data, a Strong fit cluster would receive

$$R_k = \min(R_{\min}(c_k) + 1, R_{\max}(c_k)),$$

whereas a non-Strong-fit cluster receives

$$R_k = R_{\min}(c_k).$$

For the Archival category, $R_{\min} = 0$ denotes the RS(6,3) erasure-coding baseline. An archival Strong fit, when supported by at least three clusters in the category and nonzero confidence MAD, corresponds in the offline redundancy model to RS(6,3) plus one additional full copy.

0.0.5 Step 5: Locality-Driven Relocation for a Shared File

The Locality Manager acts on files labeled as *Shared Multi-User Data*. In this example, f_3 belongs to cluster C_2 , which is assigned to the Shared category. Its replication factor remains three.

Assume that the three current replicas of f_3 are stored on DataNodes n_1 , n_2 , and n_3 . During the same observation window, the per-node demand counters $d_{f_3}(n)$ are those shown in Table 10. The demand values sum to $\text{freq}(f_3) = 760$, and the accesses originating from the three current replica nodes sum to 228, consistent with the local-access count used earlier.

Table 10: Per-node access counters for shared file f_3

DataNode n	Replica present	$d_{f_3}(n)$ accesses	Role in the move
n_1	Yes	30	Candidate source
n_2	Yes	100	Existing replica
n_3	Yes	98	Existing replica
n_4	No	532	Candidate destination

The source is the current replica with the smallest local demand:

$$s^* = \arg \min_{s \in \mathcal{R}_{f_3}} d_{f_3}(s) = n_1.$$

The destination is the node without a replica that has the largest demand:

$$d^* = \arg \max_{d \notin \mathcal{R}_{f_3}} d_{f_3}(d) = n_4.$$

The projected locality gain is

$$\begin{aligned} \Delta_{\text{loc}}(f_3; s^* \rightarrow d^*) &= \frac{d_{f_3}(d^*) - d_{f_3}(s^*)}{\text{freq}(f_3)} \\ &= \frac{532 - 30}{760} \\ &\approx 0.661. \end{aligned}$$

If this gain exceeds the configured threshold γ and the rack-diversity, capacity, node-health, and move-budget checks pass, CLEAR performs a two-phase relocation. A temporary replica is first created on n_4 , synchronization is verified, and the original replica on n_1 is then deleted. The replication factor remains equal to three, while demand originating from n_4 can subsequently be served locally.